

## Inputs

| Parameter                    | Value  | Unit               | Source                                      |
|------------------------------|--------|--------------------|---------------------------------------------|
| Positive electrode thickness | 98     | μm                 | params_lnmo_r6.json                         |
| Negative electrode thickness | 180    | μm                 | params_lnmo_r6.json                         |
| Separator thickness          | 9      | μm                 | params_lnmo_r6.json                         |
| Positive CC thickness (Al)   | 8      | μm                 | params_lnmo_r6.json                         |
| Negative CC thickness (Cu)   | 5      | μm                 | params_lnmo_r6.json                         |
| Positive porosity            | 0.3    | —                  | params_lnmo_r6.json                         |
| Negative porosity            | 0.32   | —                  | params_lnmo_r6.json                         |
| Separator porosity           | 0.55   | —                  | params_lnmo_r6.json                         |
| Electrolyte conductivity     | 6      | S/m                | params_lnmo_r6.json (estimate, LiFSI-class) |
| Cation transference number   | 0.7    | —                  | params_lnmo_r6.json                         |
| Electrolyte diffusivity      | 1e-09  | m <sup>2</sup> /s  | params_lnmo_r6.json (estimate)              |
| Cooling h                    | 100    | W/m <sup>2</sup> K | params_lnmo_r6.json                         |
| Positive particle radius     | 2.5    | μm                 | params_lnmo_r6.json                         |
| Negative particle radius     | 1.5    | μm                 | params_lnmo_r6.json                         |
| Cell area                    | 0.1027 | m <sup>2</sup>     | calc-energy area_m2                         |

## CapacityEnergy

| Quantity                   | Value             | Unit             | Formula                | Source                                             |
|----------------------------|-------------------|------------------|------------------------|----------------------------------------------------|
| Nominal capacity           | 4.5               | Ah               | parameter set          | LNMO.json                                          |
| 1C capacity (DFN)          | 8.262855276010617 | Ah               | simulation             | cell/r9_lnmo_r6_1c_dfn.json:capacity_ah            |
| Discharge energy (1C, DFN) | 35.16150824706578 | Wh               | $\int V \cdot I \, dt$ | cell/r9_lnmo_r6_energy_dfn.json:energy_wh          |
| Midpoint voltage           | 4.235000277270111 | V                | E/cap                  | cell/r9_lnmo_r6_energy_dfn.json:midpoint_voltage_v |
| 4C current (nominal basis) | 18                | A                | 4×4.5 Ah               | LNMO.json nominal capacity                         |
| 4C current density         | 175.267770204479  | A/m <sup>2</sup> | I/area                 | derived                                            |

## EnergyDensity

| Quantity                     | Value             | Unit  | Formula                                                                                   | Source                                               |
|------------------------------|-------------------|-------|-------------------------------------------------------------------------------------------|------------------------------------------------------|
| Cell mass (contract)         | 58.81265955499999 | g     | $\Sigma L \times (1 - \epsilon) \times \rho \times \text{area}$<br>(electrolyte excluded) | cell/r9_lnmo_r6_energy_dfn.json:mass_kg              |
| Energy density (gravimetric) | 597.8561165761208 | Wh/kg | E/mass                                                                                    | cell/r9_lnmo_r6_energy_dfn.json:energy_density_wh_kg |
| Volume                       | 30.81000000000001 | mL    | thickness×area                                                                            | cell/r9_lnmo_r6_energy_dfn.json:volume_m3            |
| Energy density (volumetric)  | 1141.236879164744 | Wh/L  | E/volume                                                                                  | cell/r9_lnmo_r6_energy_dfn.json:energy_density_wh_l  |
| DC resistance                | 7.98922979901996  | mΩ    | formula<br>(calc-energy)                                                                  | cell/r9_lnmo_r6_energy_dfn.json:dcr_ohm              |
| Power density (formula)      | 9.752903813752802 | kW/kg | $V^2/4R/m$                                                                                | cell/r9_lnmo_r6_energy_dfn.json:power_density_w_kg   |
| ED incl. electrolyte (info)  | 501.2719368007572 | Wh/kg | E/(mass+electrolyte)                                                                      | derived; electrolyte ρ 1.2 g/cm <sup>3</sup> lit.    |

## NPAndMass

| Quantity                   | Value                               | Unit             | Formula                                                                                                                   | Source                                                |
|----------------------------|-------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Positive areal mass        | 301.8399999999999                   | g/m <sup>2</sup> | $L \times (1 - \epsilon) \times \rho$                                                                                     | calc-energy<br>layer_kg_m2                            |
| Negative areal mass        | 202.8168                            | g/m <sup>2</sup> | $L \times (1 - \epsilon) \times \rho$                                                                                     | calc-energy<br>layer_kg_m2                            |
| N/P ratio                  | 1.922984225609263                   | —                | $(m_{\text{neg}} \cdot a_{\text{neg}} \cdot q_{\text{neg}}) / (m_{\text{pos}} \cdot a_{\text{pos}} \cdot q_{\text{pos}})$ | a 0.75/0.665 param<br>sets; q 372/146.6<br>mAh/g lit. |
| Layer masses               | 31.0 / 20.8 / 2.22 /<br>4.60 / 0.17 | g                | areal mass×area                                                                                                           | calc-energy<br>layer_kg_m2×area                       |
| Total mass<br>(contract)   | 58.812659555                        | g                | $\Sigma$ layers                                                                                                           | calc-energy<br>mass_kg                                |
| Electrolyte mass<br>(info) | 11.331918                           | g                | pore vol×1.2 g/cm <sup>3</sup>                                                                                            | pore vol =<br>area× $\Sigma(L \cdot \epsilon)$        |

## ProcessParams

| Parameter                      | Value                                | Unit              | Formula                                | Source                                                  |
|--------------------------------|--------------------------------------|-------------------|----------------------------------------|---------------------------------------------------------|
| Positive areal density         | 301.8399999999999                    | g/m <sup>2</sup>  | $L \times (1 - \epsilon) \times \rho$  | calc-energy                                             |
| Negative areal density         | 202.8168                             | g/m <sup>2</sup>  | $L \times (1 - \epsilon) \times \rho$  | calc-energy                                             |
| Positive compaction<br>density | 3.08                                 | g/cm <sup>3</sup> | $\rho \times (1 - \epsilon) \div 1000$ | derived                                                 |
| Negative compaction<br>density | 1.12676                              | g/cm <sup>3</sup> | $\rho \times (1 - \epsilon) \div 1000$ | derived                                                 |
| Electrolyte fill               | 9.443265                             | mL                | pore vol; fill factor 1.0              | derived; $\rho$ 1.2 g/cm <sup>3</sup> lit.<br>annotated |
| Formation                      | 0.1C CC to 4.7 V, 25 °C,<br>2 cycles | —                 | design recommended<br>value            | annotated:<br>production-line value<br>requires tuning  |